Trace File Analysis in NS3

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Trace File

- Tracing refers that extracting the information from the simulation results.
- Two methods to trace the data from the simulation as follows:
  - ASCII Tracing – By TraceMetrics
  - PCAP Tracing – By Tcpdump & Wireshark

ASCII Tracing

- It is similar to the 'out.tr' trace file in NS2. Add the following lines before Simulator:Run();

```
AsciiTraceHelper ascii;
pointToPoint.EnableAsciiAll(ascii.CreateFileStream
                            ("myfirst.tr"));
```

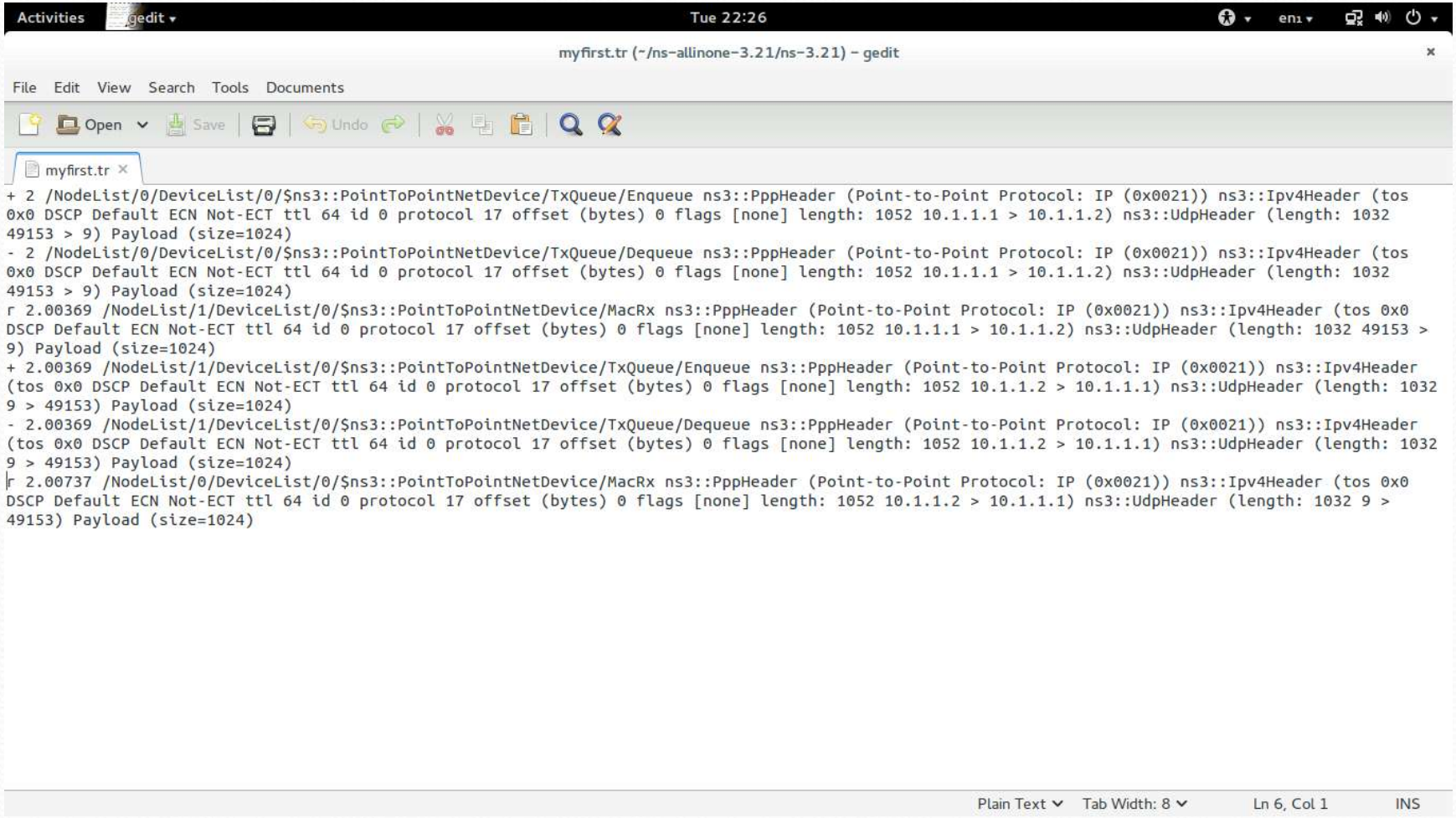
CreateFileStream → creates a file stream object on stack.

EnableAsciiAll → used to enable ASCII tracing on all P2P devices in the simulation.

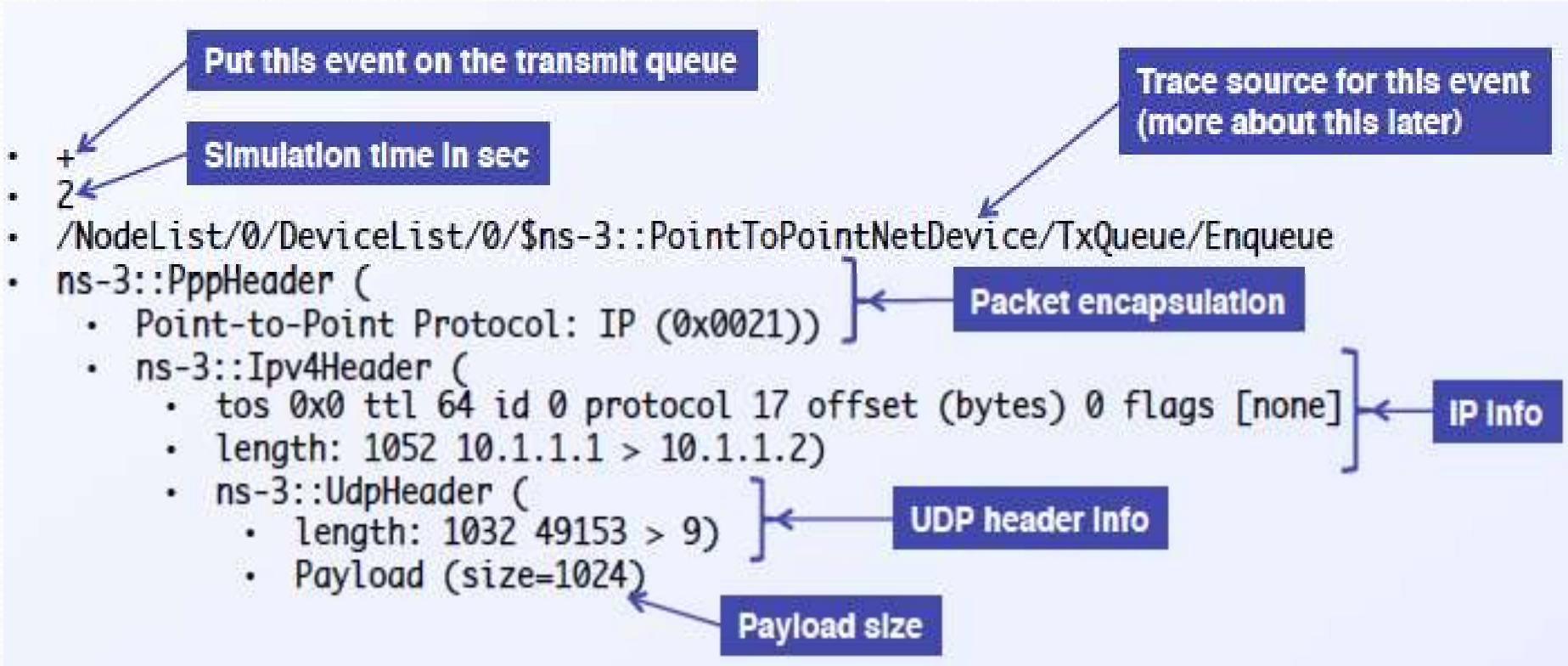
ASCII Tracing (Contd..)

- After executing the code, the trace file will be generated with the following log events:
  - + : An enqueue operation occurred on the device queue. ie., a packet is sent from one node
  - - : A dequeue operation occurred on the device queue. ie., a packet is received in the node
  - **d** : A packet was dropped, typically because the queue was full;
  - **r** : A packet was received by the net device.

ASCII Tracing (Contd..)



ASCII Tracing (Contd..)



TraceMetrics (Java Based tool)

Purpose of this tool is to perform a practical analysis of the trace file and to find the following metrics:

- No of packets sent/received/dropped
- Throughput - rate of successful data delivered over a communication channel.
- Goodput - number of useful information bits delivered by the network to a certain destination per unit of time
- Lambda - average number of packets send by the node each second.
- E[N] - average size of the queue
- E[W] - average time spent by each packet in the node
- TCP Streams – information about the delay; which includes Sequence Number, ACK, Delay, PDV(Packet Delay Variation) and IPDV (IP Packet Delay Variation (or) Jitter)
- UDP Streams – information about the packet loss

TraceMetrics

- **Prerequisites for working with TraceMetrics:**
  - JDK 7 (sudo apt-get install openjdk-7-jdk)
  - Gnuplot (sudo apt-get install gnuplot)

- Download TraceMetrics from the following link:

  http://sourceforge.net/projects/tracemetrics/

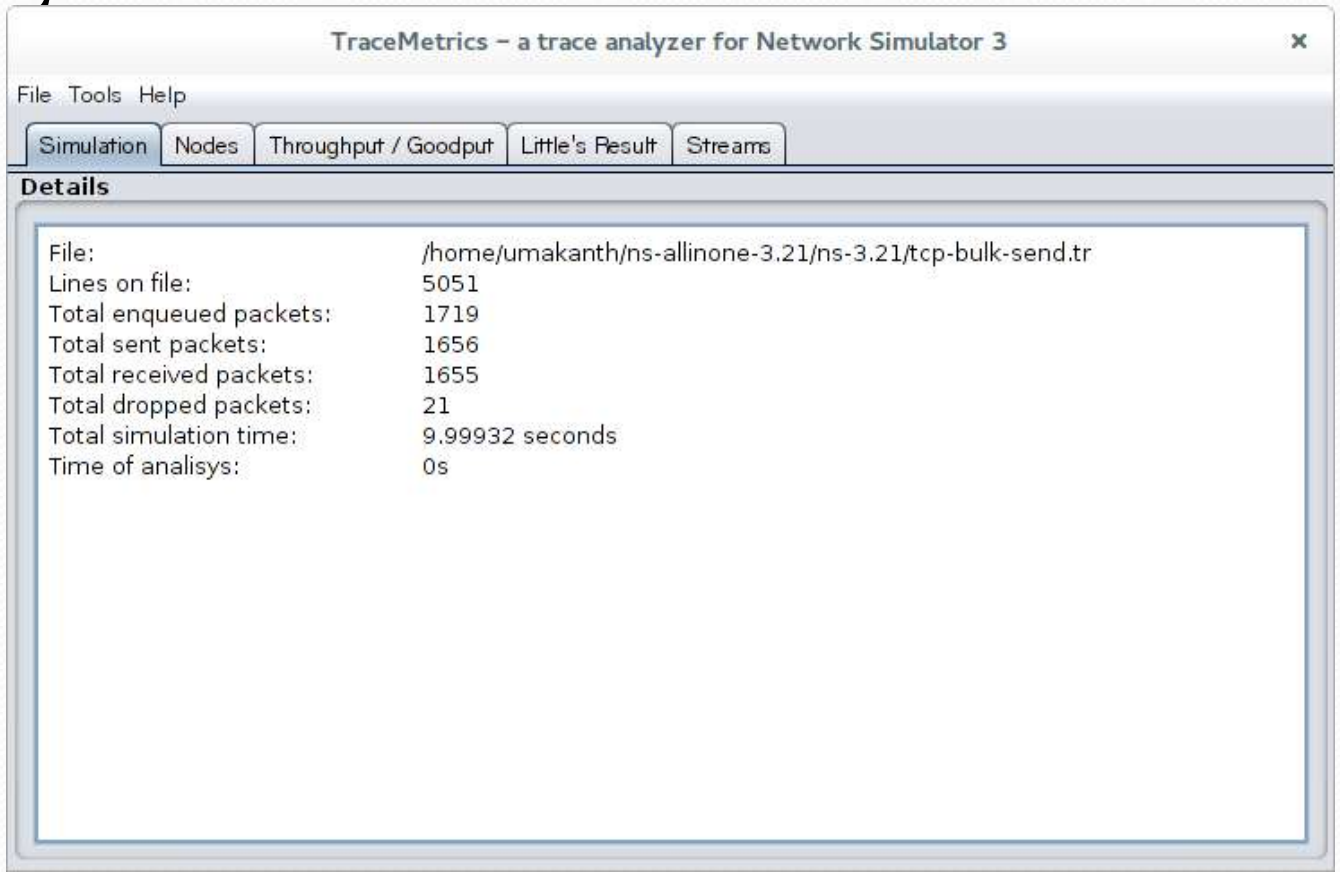
TraceMetrics (Contd..)

- Unzip the downloaded file “tracemetrics-1.4.0.zip”
- Open the TraceMetrics ($ java -jar "tracemetrics.jar”)



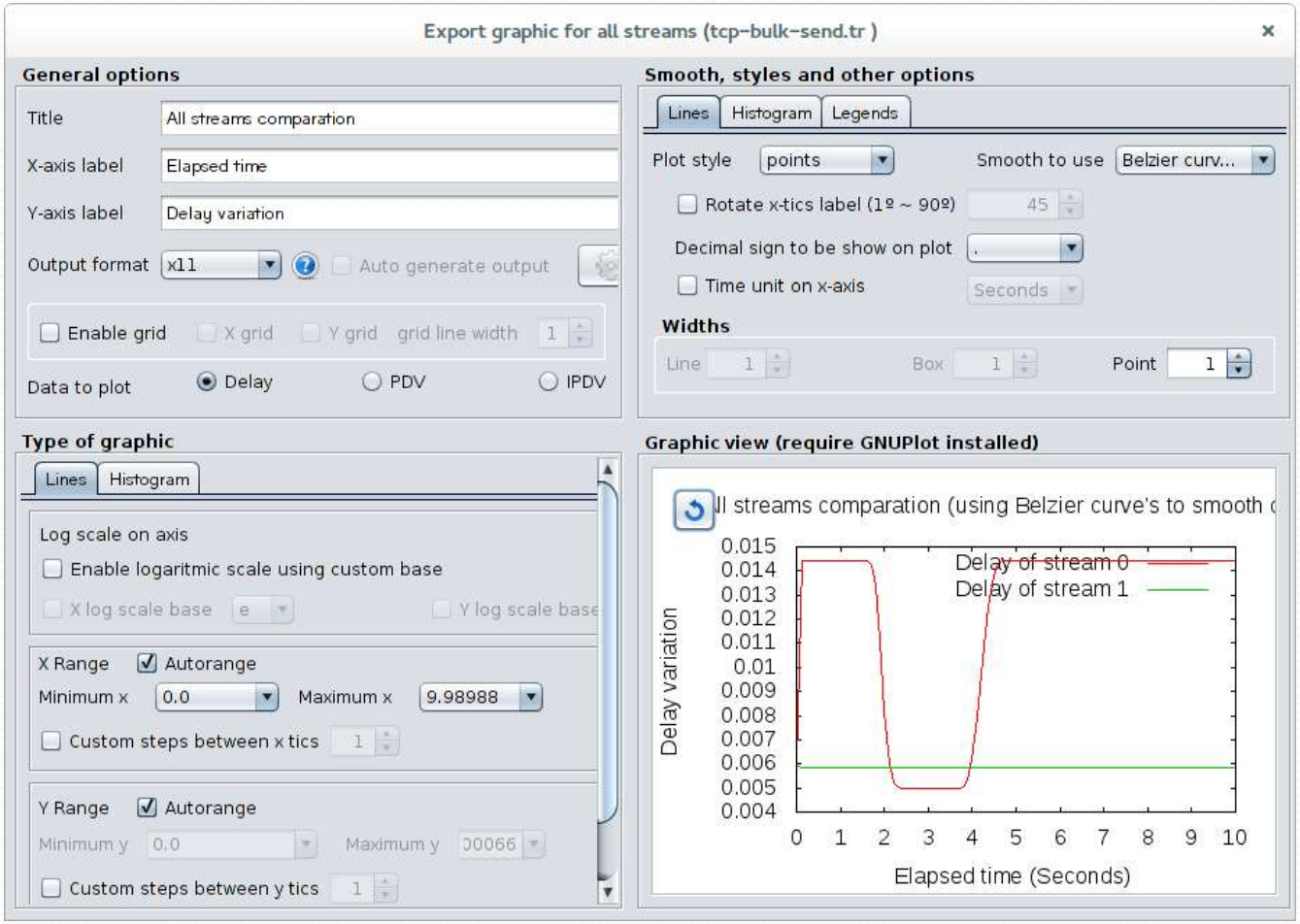
TraceMetrics (Contd..)

- Choose the TraceFile from the file menu and Click Execute Analysis button, which produces the following analysis window:



TraceMetrics (Contd..)

- We can visualize the trace file using TraceMetrics as follows:



PCAP Tracing

- PCAP refers Packet Capture; whose file extension is .pcap.
- The code used to enable pcap tracing is,

```
pointToPoint.EnablePcapAll ("myfirst");
```

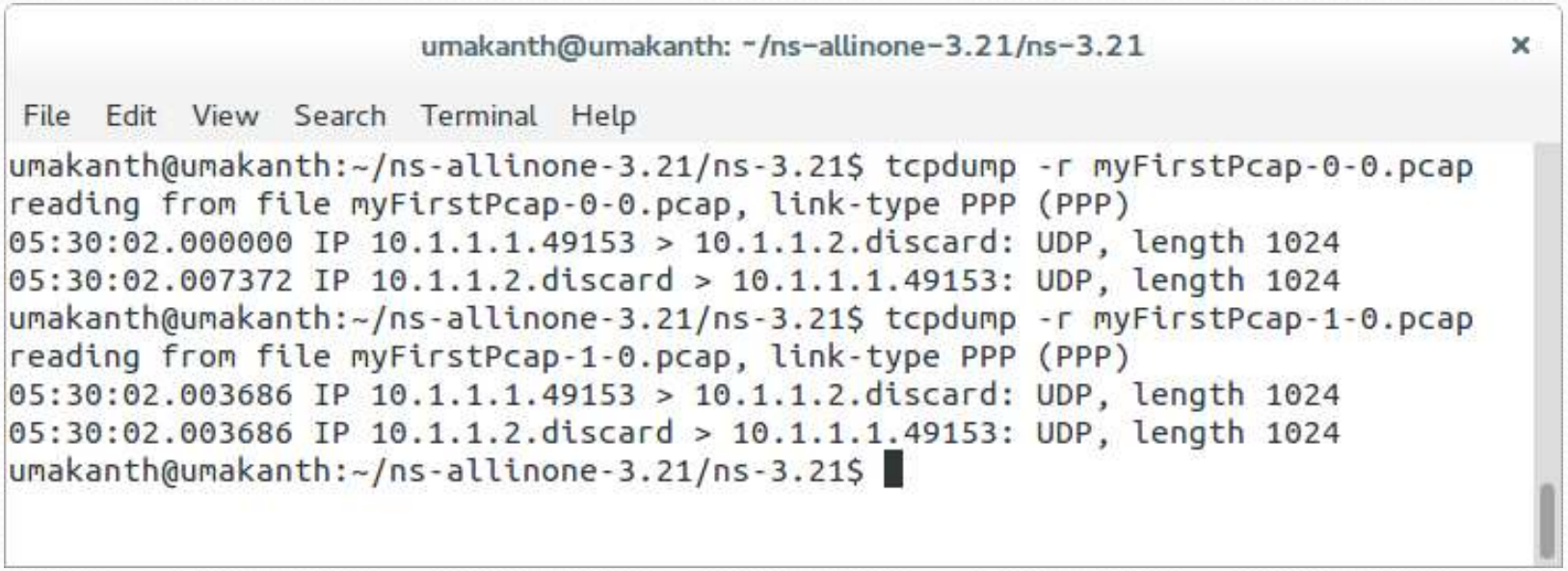
- While execution, above code will generate the .pcap file for all the devices installed in each nodes.
- For example, scenario has 1 device is installed in 2 nodes; then myfirst-0-0.pcap & myfirst-1-0.pcap will be generated.
- .pcap file can be analyzed by either Tcpdump or Wireshark

PCAP Tracing (Contd..)

- **tcpdump:**
  - Command line packet analyzer ; in which .pcap file can be read by,

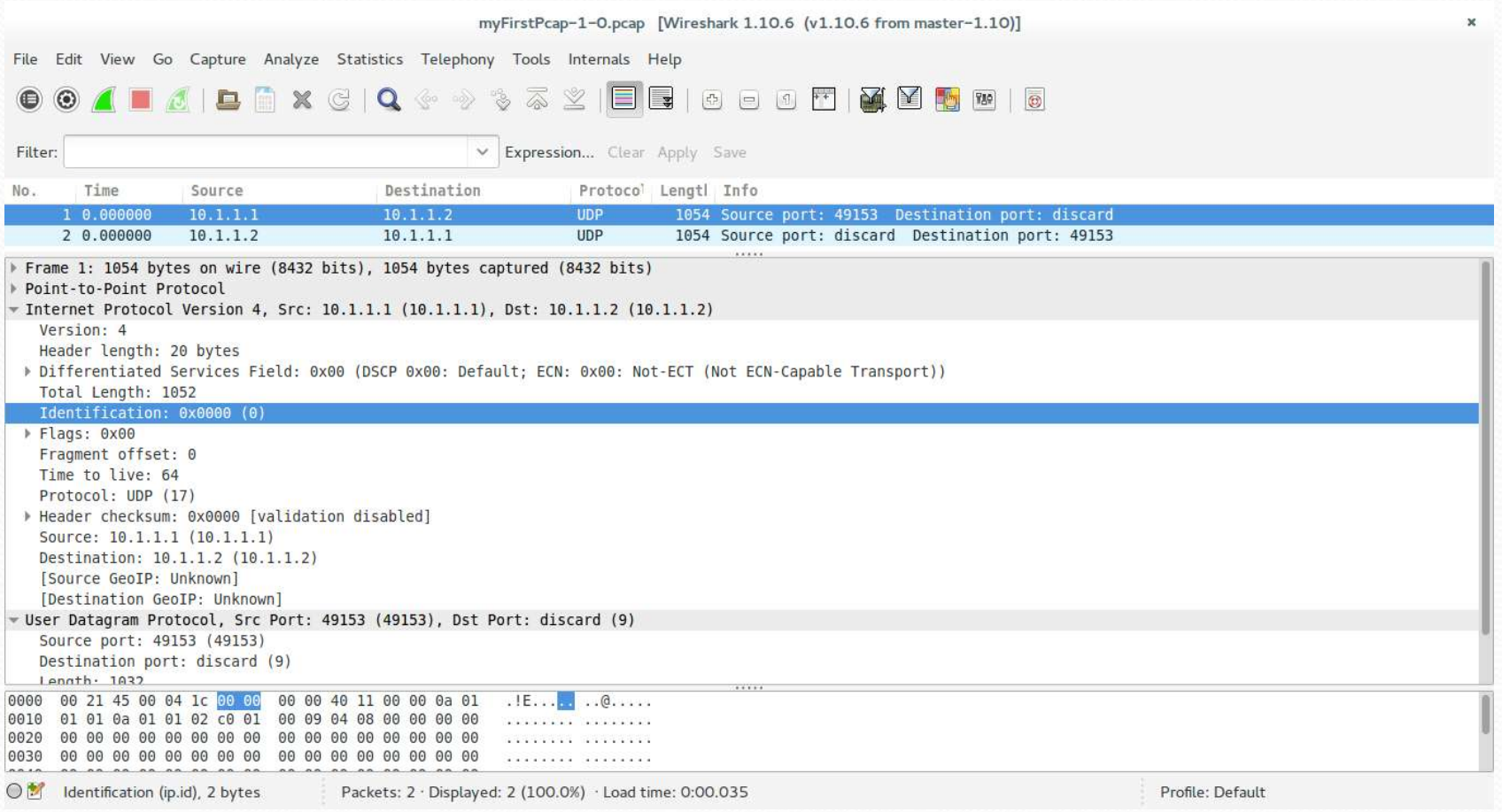
tcpdump -r myFirst-0-0.pcap //for node 0 & device 0

tcpdump -r myFirst-1-0.pcap //for node 1 & device 0



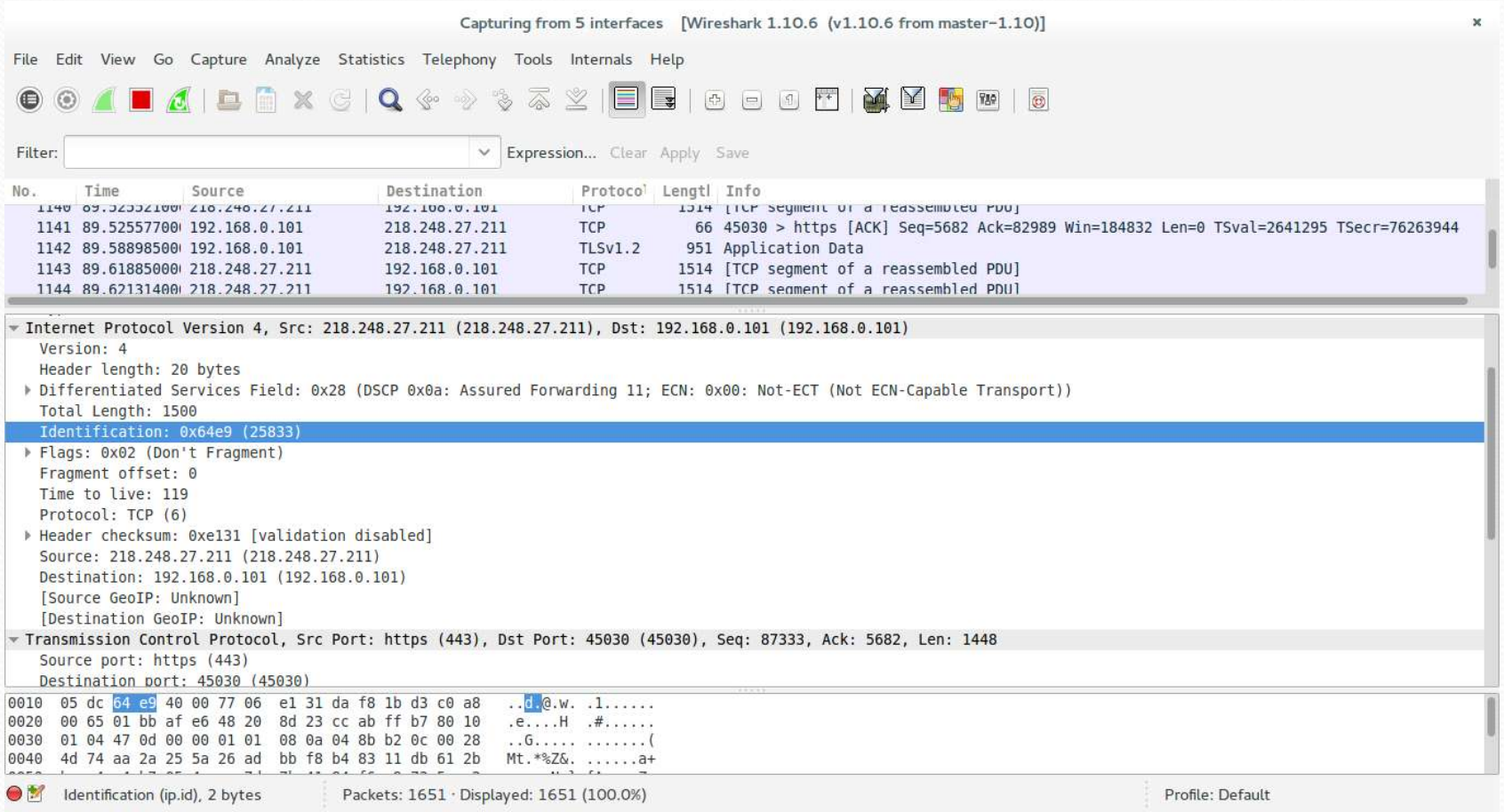
PCAP Tracing (Contd..)

- **Wireshark:** network protocol analyzer; which will read and display the .pcap file format.



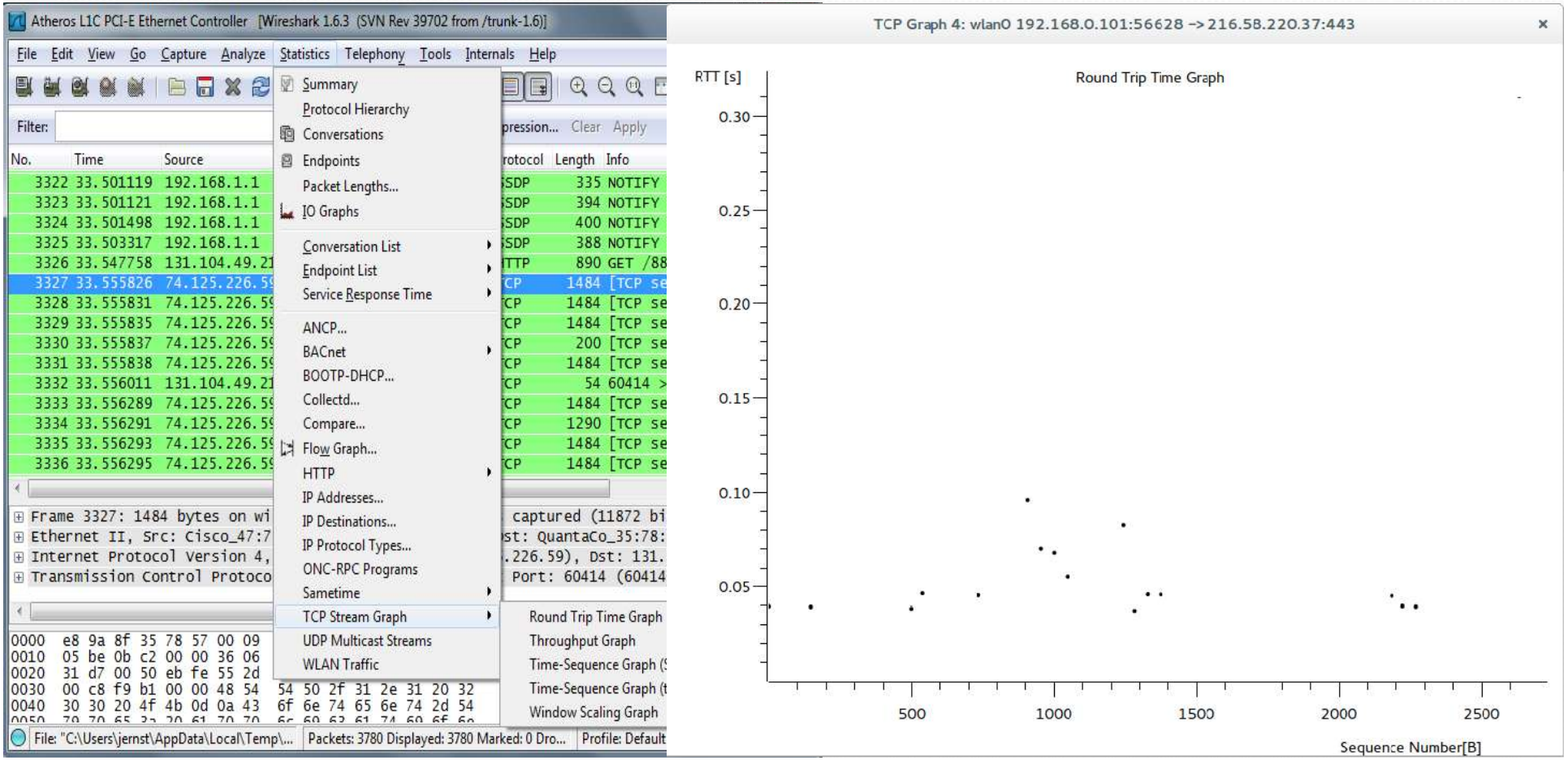
PCAP Tracing (Contd..)

- Wireshark can capture the packet transferred through our browser.



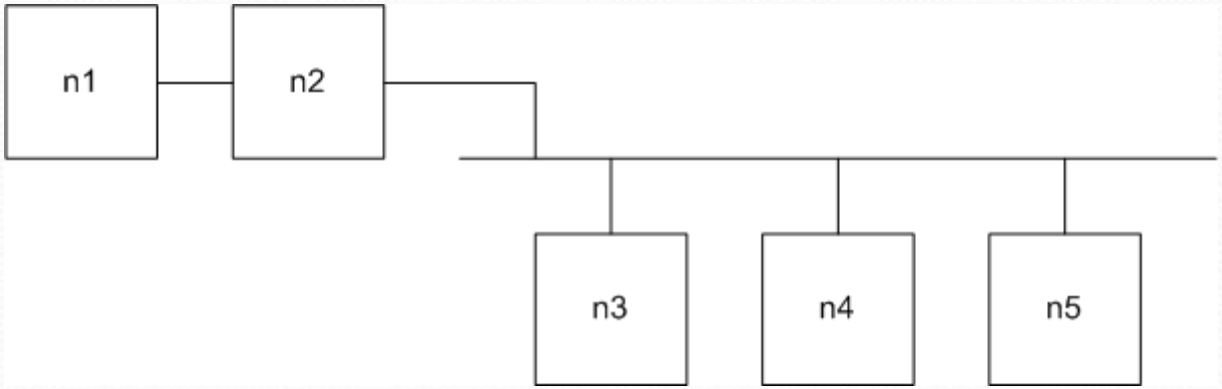
PCAP Tracing (Contd..)

- Wireshark can analyze and draw the the graph by, Statistics -> TCP Stream Graph -> Round Trip Time.



Hands-on…

- Create the following network topology and set the data rate for NetDevice is 200Kbps and packet Delay is 2ms. Also analyze the performance by both pcap and ascii trace file concepts.



- Change the metrics data rate, delay, etc and analyze the performance.

THANK YOU